

4	(a)	Two images are just resolved or distinguishable when the central maximum of one diffraction pattern coincides with the first minimum of the diffraction pattern of the other.		B1 B1
	(b)	Applying Rayleigh criterion, $\theta = \frac{\lambda}{b} = \frac{600 \times 10^{-9}}{20.0 \times 10^{-2}} = 3.00 \times 10^{-6} \text{ rad}$ $d = 8.14 \times 10^{16} \times \tan(3.00 \times 10^{-6}) = 4.26 \times 10^9 \text{ m}$		M1 A1
	(c)	(i)	For second order maxima, $d \sin \theta = 2\lambda \Rightarrow \theta = \sin^{-1} \left(\frac{2\lambda}{d} \right)$ when $\lambda = 400 \text{ nm}$, $\theta = \sin^{-1} \frac{2(400 \times 10^{-9})}{10^{-3} \div 500} = 23.578^\circ$ when $\lambda = 700 \text{ nm}$, $\theta = \sin^{-1} \frac{2(700 \times 10^{-9})}{10^{-3} \div 500} = 48.590^\circ$ Angular spread $\Delta\theta = 48.590^\circ - 23.578^\circ = 25.0^\circ$	M1 M1 A1
		(ii)	Advantage: θ is larger, so for the same $\Delta\theta$, the fractional uncertainty / percentage uncertainty in θ is smaller.	B1
			Total:	8

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C2

5	(a)	(i)	Resistance of thermistor is given by the ratio of p.d. across thermistor V to	R_1
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